

Exploring the Correlation between Cross-Lingual Alignment and the Generative Capabilities of Large Language Models: Dataset and Problem Approval

1. Literature Review

The State and Fate of Linguistic Diversity and Inclusion in the NLP World

Joshi et al., 2020

Read by Sasha Dmitriev

This paper analyses an aggregate of many computational linguistics journals and conferences to find which languages have historically and currently been underrepresented in the development of LLMs. It also used the number of Wikipedia articles as a measure of unlabeled data, classifying languages into 6 different categories:

0. The Left-Behinds - Languages that are still being ignored, both on the internet and by researchers.
1. The Scraping-Bys - Languages with some unlabeled data, but very little to no labeled data.
2. The Hopefuls - Languages with some unlabeled data, and a small set of labeled data.
3. The Rising Stars - Languages with strong unlabeled data, but not enough labeled data.
4. The Underdogs - Languages with large amounts of both labeled and unlabeled data.
5. The Winners - Languages with the highest amount of resources. These languages profit the most from breakthroughs in machine learning.

In the Entity Embedding Analysis, a vector diagram was created, plotting conferences, authors, and languages against each other. Understanding the visualization was the most challenging part of the paper. It claims that some authors who gravitate towards certain conferences also gravitate towards certain languages, which can explain the differences in language inclusivity between different conferences.

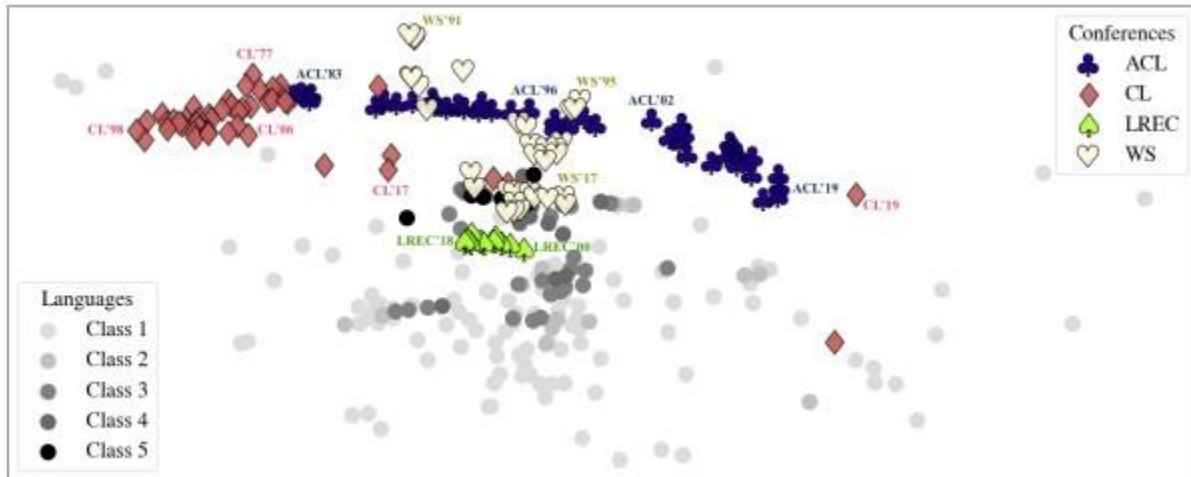


Figure 1: t-SNE visualization of the learned conference and language embeddings (from Joshi et al., 2020).

An important finding was that languages with more ignored (less common) typological features were associated with worse performance when working in those languages, compared to languages with features more common in high-resource languages. This is an important assumption for the study we are reproducing, since we are attempting to improve the performance of LLMs in said languages.

Do Explicit Alignments Robustly Improve Multilingual Encoders?

Wu & Dredze, 2020

Read by Sasha Dmitriev

The paper aimed to answer the question “Does explicit alignment improve cross-lingual representations (the similarity between the vectors of words in different languages)?” I considered the languages Arabic, German, English, Spanish, French, Hindi, Russian, Vietnamese, and Chinese using Google’s mBERT and Meta’s XLMR models to compare their efficacy at zero-shot cross-lingual transfer. It used performance on XNLI(Cross-lingual Natural Language Inference), NER(Named Entity Recognition), and POS tagging(Part-of-Speech Tagging) tasks to compare different forms of explicit alignment.

	XNLI	NER	POS	Parsing		XNLI	NER	POS	Parsing
mBERT	70.1 $\pm$ 0.8	67.7 $\pm$ 1.3	78.3 $\pm$ 0.5	52.6 $\pm$ 0.4	mBERT	70.1 $\pm$ 0.8	67.7 $\pm$ 1.3	78.3 $\pm$ 0.5	52.6 $\pm$ 0.4
+ Linear Mapping	70.0 $\pm$ 0.6	63.7 $\pm$ 1.5	79.5 $\pm$ 0.5	53.6 $\pm$ 0.3	+ Linear Mapping	70.2 $\pm$ 0.6	63.8 $\pm$ 1.3	80.1 $\pm$ 0.4	53.6 $\pm$ 0.3
+ L2 Align	69.7 $\pm$ 0.4	67.1 $\pm$ 1.0	78.0 $\pm$ 1.3	52.2 $\pm$ 0.7	+ L2 Align	70.3 $\pm$ 0.5	67.8 $\pm$ 1.4	78.2 $\pm$ 1.2	52.8 $\pm$ 0.7
+ Weak Align (Our)	70.5 $\pm$ 0.7	68.0 $\pm$ 1.3	78.8 $\pm$ 0.7	53.1 $\pm$ 0.6	+ Weak Align (Our)	70.8 $\pm$ 0.7	67.3 $\pm$ 0.9	78.8 $\pm$ 0.6	52.9 $\pm$ 0.6
+ Strong Align (Our)	70.4 $\pm$ 0.7	67.7 $\pm$ 1.1	79.0 $\pm$ 0.7	53.0 $\pm$ 0.6	+ Strong Align (Our)	70.4 $\pm$ 0.7	67.2 $\pm$ 1.1	79.0 $\pm$ 0.7	53.3 $\pm$ 0.6
XLMR _{base}	76.4 $\pm$ 0.5	66.4 $\pm$ 0.9	81.2 $\pm$ 0.6	57.3 $\pm$ 0.6	XLMR _{base}	76.4 $\pm$ 0.5	66.4 $\pm$ 0.9	81.2 $\pm$ 0.6	57.3 $\pm$ 0.6
+ Linear Mapping	73.4 $\pm$ 0.6	54.1 $\pm$ 0.9	81.3 $\pm$ 0.5	55.6 $\pm$ 0.5	+ Linear Mapping	73.5 $\pm$ 0.5	54.2 $\pm$ 0.8	81.7 $\pm$ 0.6	56.1 $\pm$ 0.4
+ L2 Align	75.7 $\pm$ 0.5	65.7 $\pm$ 1.2	81.3 $\pm$ 0.9	56.2 $\pm$ 0.7	+ L2 Align	75.8 $\pm$ 0.5	65.5 $\pm$ 1.2	81.4 $\pm$ 0.8	55.9 $\pm$ 0.6
+ Weak Align (Our)	76.1 $\pm$ 0.7	66.0 $\pm$ 1.0	81.5 $\pm$ 0.5	57.4 $\pm$ 0.4	+ Weak Align (Our)	76.0 $\pm$ 0.4	66.2 $\pm$ 1.2	81.5 $\pm$ 0.5	57.4 $\pm$ 0.5
+ Strong Align (Our)	76.0 $\pm$ 0.6	66.1 $\pm$ 0.9	81.4 $\pm$ 0.6	57.4 $\pm$ 0.5	+ Strong Align (Our)	76.1 $\pm$ 0.4	66.2 $\pm$ 1.0	81.5 $\pm$ 0.6	57.4 $\pm$ 0.5
XLMR _{large}	80.4 $\pm$ 0.6	71.0 $\pm$ 1.4	82.6 $\pm$ 0.5	59.4 $\pm$ 0.8	XLMR _{large}	80.4 $\pm$ 0.6	71.0 $\pm$ 1.4	82.6 $\pm$ 0.5	59.4 $\pm$ 0.8

(a) Alignment with bitext used in previous works

(b) Alignment with the OPUS-100 bitext

Table 1: Zero-shot cross-lingual transfer result, average over 9 languages (from Wu & Dredze, 2020).

To adequately replicate the findings of Ravisankar et al., it is important to understand the different forms of alignment and their impact on model performance. However, it is equally important to understand how each method actually weights the data. Deepening our understanding of linear mapping, L2 alignment, and weak and strong alignment is imperative for understanding our own findings.

Understanding Cross-Lingual Alignment—A Survey

Hämmerl et al., 2024

Read by An Le

This paper provides valuable background on cross-lingual alignment through a comprehensive literature survey, which is crucial for evaluating how multilingual models align representations across languages: [Understanding Cross-Lingual Alignment—A Survey](#).

The research in the paper focuses on 4 main points:

1. The definition of “Alignment” in cross-lingual alignment is presented in 2 views:
 - a. Presentation similarity (Weak alignment and strong alignment)
 - i. How hard is it to get strong alignment
 - b. How the prediction head utilized the embedded space
 - i. Split the embedded space into 2 subspaces: Language-neutral subspace and Language-sensitive subspace
 - ii. A classification head attached to the model doesn't need every dimension to align; it just needs a linear projection in which the task-relevant signal dominates language noise.

2. Measuring Cross-Lingual Alignment
 - a. Cosine similarity
 - b. Word or Sentence Retrieval Tasks
 - c. Zero-Shot Transfer After Fine-Tuning
 - d. Language identification
 - e. Visualization
3. Taxonomy of Alignment Techniques
 - The paper group techniques are based on their training objectives, initialization methods, and data requirements.
4. Limitation (content for future research)
 - a. Bilingual vs Multilingual: the dominance of English → the need for research on the translation of different languages
 - b. Multimodality

The obstacles:

There are several obstacles due to the limitation of knowledge about the transformer model and the math behind.

1. Contextual embedding and prediction head (resolved)
 - How the contextual embedding works
 - Where does the prediction head live
2. Encoder-only and Decoder-only models (resolved)

	Encoder-only	Decoder-only
Attention	Bidirectional	Causal / left-to-right
Main purpose	Understanding/representation	Generation
Example	BERT	GPT
Typical training objective	Masked Language Modeling	Next-token prediction
Output	Contextual representations	Generated sequence

This is the necessary information to understand the BERT, mBERT, and XLM-R, and also state the need for research for a decoded-only model (encoder model that is used for the improvement technique of the cross alignment in the paper)

3. Anisotropy and Hubness for understanding why representations from two languages can be difficult to separate or align cleanly, and how the contrastive learning has been shown to make pretrained embedding spaces more uniform while improving alignment of positive pairs (in-progress)

MEXA: Multilingual Evaluation of English-Centric LLMs via Cross-Lingual Alignment

Kargaran et al., 2024

Read by Bav Sandhu

The most challenging part was understanding how the MEXA score is calculated from the similarity matrix. Also, selecting different layers changed the score, making it difficult to learn. Establishing if alignment actually causes better performance was another obstacle since the paper only shows that alignment with performance goes up together.

Online Language Adaptive Sampling for Better Distributed Cross-lingual Gains

Nguyen et al., 2026

Read by Bav Sandhu

The most challenging part was following the two-loop setup and the UCB formula used to update the language weights. The formulas deciding the amount by which a language is selected was hard to follow, when training more on poorly aligned languages.

Middle-Layer Representation Alignment for Cross-Lingual Transfer in Fine-Tuned LLMs

Liu et al., 2025

Read by Matthew MacDonald

This paper provides insight on potential methods for improving cross lingual alignment in decoder only large language models, in this case specifically Qwen-2.5, and Llama-3. To test cross lingual alignment, the FloRes-200 dataset was used to see if representations of sentences from different languages were similar to each other. The models were then prompted to identify a correct translation from language to language and through these methods it was found cross lingual alignment is strongest in the middle layers of decoder models. The models were then tasked with performing Slot filling (on MASSIVE and Tatoeba dataset), Machine translation (on WMT 23 and ALMA dataset and measured with BLEU, and COMET-22), and producing JSON formatted bodies (on UNER and Taoeba dataset).

	Dataset	Languages
Slot Filling		
Task - train	MASSIVE	{ar, en, es, ru, zh}
Task - test	MASSIVE	supervised + {af, az, cy, de, el, fr, hi, is, ja, jv, sw, th, tl, tr, ur}
Alignment	Tatoeba	low-res.: {cy, jv, jp, sw, tl}-en mid-res.: {el, hi, th, tr}-en high-res.: {ar, es, ru, zh}-en
Machine Translation		
Task - train	ALMA	{cs, de, is, ru, zh} ↔ en
Task - test	WMT 23	supervised + {he, ja, uk} ↔ en
Alignment		(same as “Task - train”)
JSON Generation (challenge task)		
Task - train	UNER	{en, pt, zh}
Task - test	UNER	supervised + {da, hr, sk, sr, sv}
Alignment	Tatoeba	{da, sv}-en
Semantic Alignment Evaluation (diagnostic task)		
Alignment	FLoRes-200	$N(N - 1)$ pairs for N lang.

Table 1: Dataset statistics for three downstream tasks and one diagnostic task. “Train” refers to languages involved in SFT, and "test" includes SFT languages and additional transfer languages unseen during training.

When the models were fine-tuned by specific task training and using contrastive loss it was observed that alignment between related semantic content improved, and unrelated content were pushed farther from each other, these methods were applied to the models middle layers. The benefits from this training technique were found to benefit not just the languages used for training but also improved transfer to other languages. It was also found that these benefits did not fully transfer to non latin scripts.

How do Large Language Models Handle Multilingualism?

Zhao et al., 2024

Read by Matthew MacDonald

This paper examines how large language models internally handle multilingual inputs. The authors hypothesize that large language models follow a specific workflow. Firstly the model processes the language in which it is prompted and moves its representation closer to an English centric one. The model then reasons and problem solves in this English centric representation, then in the later layers the model transfers its english result back to the original prompted language. Their hypothesis was tested through testing Vicuna and Mistral large language models capabilities around understanding(XQuAD), reasoning(MGSM), knowledge(X-CSQA), and answering(XLSum) whilst deactivating language related neurons. By deactivating target neurons, they found that after deactivating a very small portion of the model's neurons it was found that the model's capabilities for performing multilingual tasks was substantially negatively impacted

for certain languages and tasks.

Model	Deactivating Method					Performance				
	Under	S-ATTN	S-FFN	Gen	Neuron	Eng	n-Eng	Δ_{Eng}	$\Delta_{\text{n-Eng}}$	$\Delta \uparrow$
Vicuna	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	Random	57.8	53.9	+0.3	-0.1	+0.4
	$\times$	$\times$	$\times$	$\times$	Random	57.9	54.2	+0.4	+0.3	+0.1
	$\checkmark$	$\times$	$\times$	$\checkmark$	Lang-Spec	40.9	38.6	-15.9	-15.3	-0.6
	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	Lang-Spec	57.9	52.8	-0.4	-1.1	+0.7
	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	Lang-Spec	56.5	46.0	-0.5	-7.9	+7.4
Mistral	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	Random	58.1	55.5	+1.0	-0.2	+1.2
	$\times$	$\times$	$\times$	$\times$	Random	57.6	55.5	+0.5	-0.2	+0.7
	$\checkmark$	$\times$	$\times$	$\checkmark$	Lang-Spec	53.2	47.0	-3.9	-8.7	+4.8
	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	Lang-Spec	56.4	54.6	-0.7	-1.0	+0.3
	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	Lang-Spec	56.2	48.3	-0.9	-7.4	+6.5

2. Datasets

For the reproduction stage, we will use the multilingual benchmark datasets used in the original paper: Belebele (Bandarkar et al., 2024), Xstudycloze (Lin et al., 2022), and Xscopa (Ponti et al., 2020).

1. Belebele for reading comprehension (“Can you understand what you read?”)
 - Dataset link: https://github.com/facebookresearch/belebele?utm_source=chatgpt.com
 - Paper: Bandarkar et al. (2024)
 - Task: Multiple-choice reading comprehension
 - Languages: 122 language variants
 - Size: 109,800 questions total — 900 per language variant
 - Source: passages from FLORES-200.
2. XStoryCloze for narrative and commonsense reasoning (“Can you understand what should happen next in a story?”)
 - Dataset Link: https://github.com/facebookresearch/fairseq/blob/main/examples/xglm/XStoryCloze.md?utm_source=chatgpt.com
 - Paper: Lin et al. (2022)

- Task: Story completion / commonsense reasoning
 - Languages: English + 10 additional languages
 - Original dataset: StoryCloze
 - Purpose: Evaluate multilingual zero-shot and few-shot capabilities.
3. XCOPA for causal commonsense reasoning (“Can you understand why something happened or what it causes?”)
- Dataset link: https://github.com/cambridgeltl/xcopa?utm_source=chatgpt.com
 - Paper: Ponti et al. (2020)
 - Full name: XCOPA: A Multilingual Dataset for Causal Commonsense Reasoning
 - Task: Choose the most plausible cause or effect
 - Languages: 21 languages.
 - XCOPA is based on COPA, the Choice of Plausible Alternatives task.

License

Dataset	License
BELEBELE	CC BY-SA 4.0
XStoryCloze	CC BY-SA 4.0
XCOPA	CC BY 4.0

Data visualization

In the code. File path: /notebooks/datavis.ipynb

3. Model Selection

- Model architecture: Decoder-only LLMs.
- Model source: Models available through Hugging Face.
- Models:
 - Llama 3.2 1B Instruct — Meta
 - Hugging Face: [Llama-3.2-1B-Instruct](#)
 - Qwen3 0.6B Base — Qwen
 - Hugging Face: [Qwen3-0.6B-Base](#)
 - Gemma 3 1B IT — Google

- Hugging Face: [gemma-3-1b-it](#)
- Alternative: Approximately 3B-parameter models using quantization to reduce computational and memory requirements.

4. Data Processing Plan

1. Choose English and 12 other languages data (depends on the computing resources)
2. Replicate the way the original paper processes it:
 - a. BELEBELE: Uses the passage + question + 4 answer options. They evaluate it as a 4-choice MCQA task.
 - b. StoryCloze: Uses the 4 story sentences + 2 possible endings and treats it as a 2-choice MCQA task.
 - c. XCOPA: Uses the premise + 2 choices, with the task asking for the plausible cause/effect.
3. Shrink down the data based on the computational resources.

5. References

- Wu, S., & Dredze, M. (2020, November). Do Explicit Alignments Robustly Improve Multilingual Encoders? In B. Webber, T. Cohn, Y. He, & Y. Liu (Eds.), *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 4471–4482). doi:10.18653/v1/2020.emnlp-main.362
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